



Co-funded by
the European Union

Project number: 2023-1-
FR01-KA220-SCH-
000151881

PASS THE PARCEL

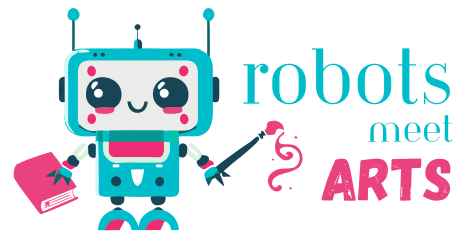


STIMMULI
for social change



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Subject: Geography

Content: Build geographical reference points, transport modes and networks, Communicate thanks to the Internet

Computational practices:
choose what fits

- Abstraction
- Algorithms
- Problem solving

Unplugged/plugged:
unplugged



Preparation time:

60 - 90 min



Age:

8-12 years



Duration:

190 min



Location:

indoor/outdoor

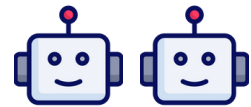


Group distribution:

full classroom/in group



Difficulty



After this lesson, children will be able to:

- Build Geographical Reference Points: Design routes using different transportation systems, considering distance, time, and environmental impact for a hands-on understanding of geography.
- Explore Transport Modes and Networks: Research various transport options, discussing their efficiency, costs, and environmental impacts, while designing routes based on unique transportation constraints for practical insights into daily commuting.
- Understand Internet Communication: Learn about routing algorithms that determine efficient data paths across networks, drawing parallels to transportation route planning to introduce concepts of internet communication.



Materials needed:

- **Colored vests, jackets or stickers** (2 of each color: red, green, blue, yellow) to identify the role of each student (representing nodes) in the simulation of the routing algorithm.
- **4 balls** corresponding to the colors of the vest to simulate the data packets circulating on the network.
- **Internet access to conduct online research** (you can also provide printed resources) **and to use mapping or navigation tools to plan routes**
- **Art materials** (optional) to create panels representing the modes of transport studied.
- **Ticket to Ride board game** (optional) to introduce the concept of routing algorithms in a fun and engaging way.
- **Maps of the studied area** on which different levels of complexity will be represented (for example using layers, pictograms, etc.).
- **An informative map of their neighborhood** to plan the route of the final activity.
- **A GPS and compass** to aid in navigation and route checking during the final activity.

Helpful links:

- Discover Games enabling to discover routing:
 - <https://www.daysofwonder.com/ticket-to-ride/>
 - <https://www.supermeeple.com/nos-jeux/expeditions/>
- Explore routing algorithms:
 - <https://www.youngwonks.com/blog/routing-algorithms>
 - https://kids.kiddle.co/Maze_solving_algorithm
- Going further with a complementary unplugged activity:
 - https://classic.csunplugged.org/documents/activities/network-protocols/unplugged-en-network_protocols-v3.1.pdf

Preparation and teacher notes

Prepare colored vests or stickers (2 of each color: red, green, blue, yellow).

Get 4 balls matching the vest/sticker colors.

Scout and select an appropriate outdoor space that is:

- Large enough for students to spread out
- Has a flat, safe surface
- Away from major distractions

Draw or print the network diagram to use as reference when positioning students.

Create a backup indoor space plan in case of bad weather, such as a gymnasium.

Prepare simple visual aids to explain network concepts.

Warm up - 1 hour

Make Children Move

To make the children actively involved in their learning, organise a first unplugged activities involving body movement and active participation.

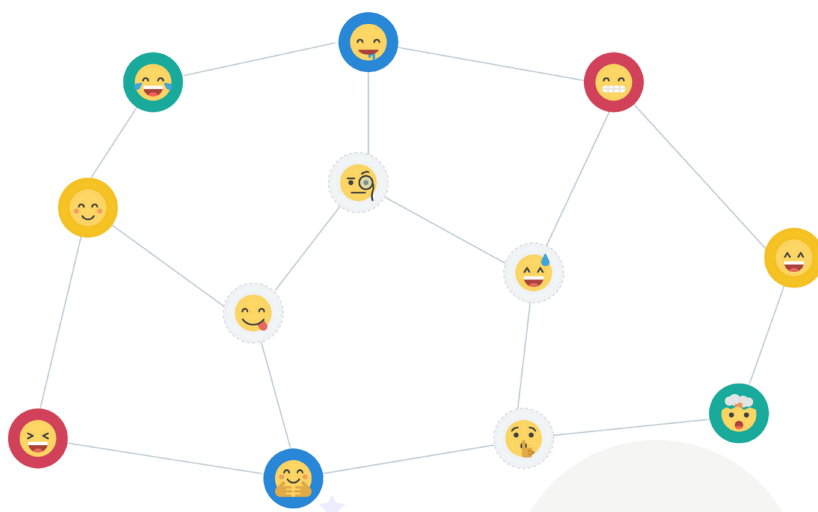
Take the class outside to simulate how computer networks route information.

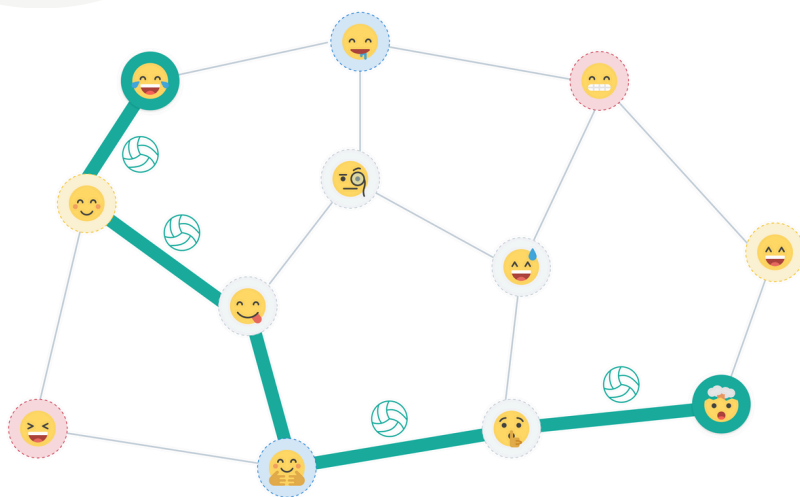
Give 8 children a colour vest (or a sticker, or whatever material you have to materialise colour), in pairs: 2 red, 2 green, 2 blue and 2 yellow for instance and prepare 4 balls of the same colours. The rest of the classroom does not receive colours.

Set up the game for all the rounds. Position the pupils according to a defined network structure (you can create yours or use the one we have given in example, just make sure that children with the same colour are far enough). They will play the role of node in a network.

A network node sits at a point in the network where it sends, receives, stores or creates information. It transmits data to communicate with other nodes in the network.

Nodes connect over a link or communication channel. In a computer network these may be cable, fiber optic or wireless connections.





Give the first round objective:

Green Vest Pupil #1 must pass the Green Ball to Green Vest Pupil #2. Pupils must pass the ball between them until it reaches its final destination without making the ball dropped on the floor, simulating how data packets travel through a network.

Since students are positioned far apart, direct passes risk dropping the ball - representing packet loss in network terms. If the balls (i.e. packet) falls, then we can consider it as a packet loss.

A **packet** is a small unit of data that a network protocol routes between an origin and a destination on the internet or any other packet-switched network. Network packets hold small amounts of data that typically include information such as the source and destination address, protocols or identification numbers. From sending emails to downloading videos, every internet activity requires the transfer of packets.

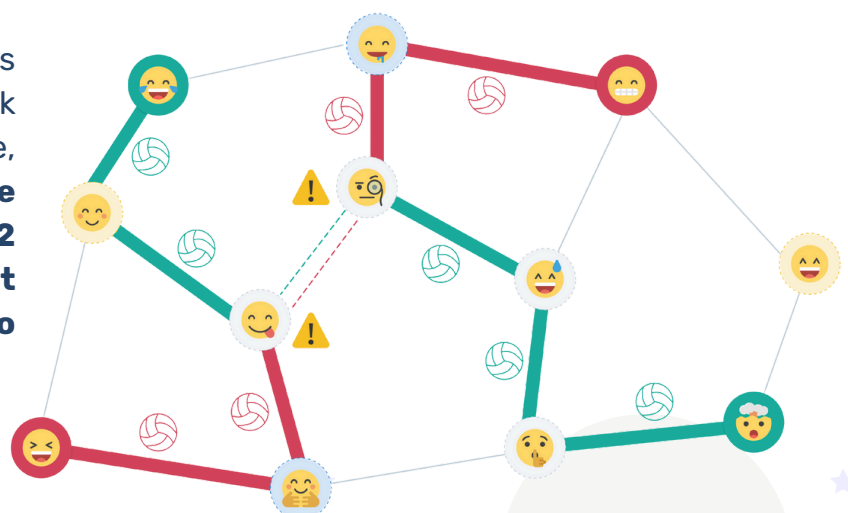
Packet loss occurs when one or more transmitted data packets fail to arrive at their destination. This can cause noticeable performance issues for all types of digital communications.

In the proposed setup, the color of the ball represent the notion of “**address**” of a packet and the message inside is the data gathered from the sender to the receiver.

This physical demonstration helps students understand network nodes, routing decisions, and potential communication issues.

To increase complexity and learning opportunities, organise additional rounds:

In **round 2**, introduce multiple balls simultaneously to show network congestion concepts. In this case, **Green Vest Pupil #1 must pass the Green Ball to Green Vest Pupil #2 while in the meantime, Red Vest Pupil #1 must pass the Red Ball to Red Vest Pupil #2.**

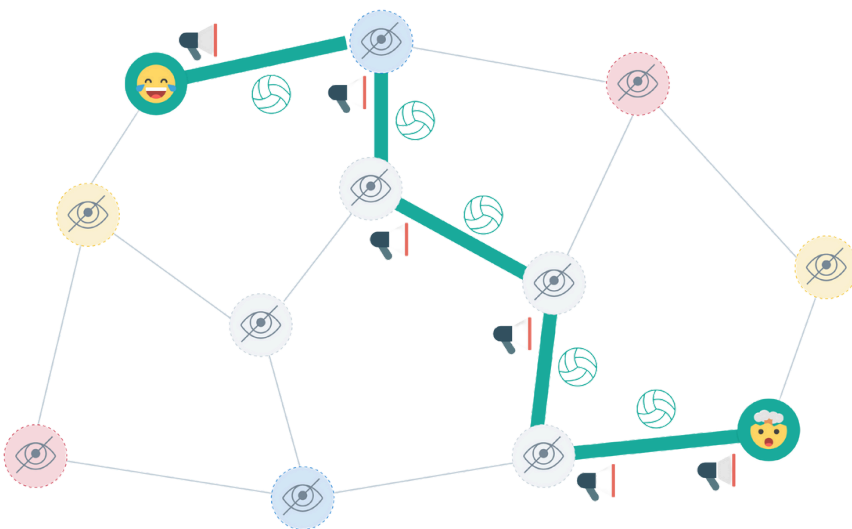


Pupils must pass the balls between them until they reach their final destinations without making the balls dropped on the floor, simulating how multiple data packets travel through a network simultaneously.

This creates a situation where some pupils (nodes) in the network might receive multiple balls (packets) simultaneously. When a pupil receives multiple balls at once, they must make a decision about which ball to pass first - this simulates how computer networks handle traffic congestion. Students can experiment with different strategies:

- **First-come-first-served:** Pass the balls in the order they were received
- **Color priority:** Establish a predetermined order (e.g., red balls have priority over blue ones)
- **Destination proximity:** Pass first the ball that is closest to its final destination

This situation mirrors real network congestion where nodes (like internet routers) must manage multiple data packets arriving simultaneously. When networks become overloaded, some packets might be delayed or even dropped, just like how students might struggle to handle multiple balls at once.



Eventually, you can optionally add a **final round** where pupils, except the ones delivering and receiving the packet, are blindfolded. For **this third round**, return to the initial setup with just one ball being passed (Green Ball from Green Vest Pupil #1 to Green Vest Pupil #2), Blindfold all pupils except the sender and final receiver.

Before blindfolding, give students a moment to memorize their positions and the positions of their neighbors. This is crucial as they will need to maintain their spatial awareness throughout the exercise.

The success of this round depends heavily on clear verbal communication and spatial memory. Students must:

- Remember who is standing to their left, right, front, and back
- Give precise instructions like "I'm passing to your left hand" or "The ball is coming from behind you"
- Listen carefully to determine where the ball is in the network

This simulates how networks must function when intermediate nodes have limited information, and demonstrates the importance of maintaining network topology awareness even when direct visual feedback is not available.

Reflect on the unplugged activity.

After completing each round or at the end of the whole unplugged activity, gather the students to discuss their experience and observations.

Draw the network map you have created with the pupils on the board of the classroom to help materialise and analyse what happened.

Discuss how each student (acting as a network node) made simple decisions that contributed to solving the complex routing challenge.

Point out key observations for each round and **question the pupil regarding how information could reach different destinations through various paths (routing) and the importance of reliable exchanges (avoiding dropped balls/lost packets).**

Start with the first round, where a single ball traveled through the network. Ask them to describe the journey of the ball and how they made decisions about where to pass it. This simple scenario helps introduce the basic concept of data routing.

When moving to the second round with multiple balls, the discussion becomes more interesting. Pupils will likely notice how the network became more challenging to manage. Some might have experienced moments of confusion when receiving multiple balls at once, mirroring how real networks handle traffic congestion.

This is an excellent opportunity to explore how their different strategies emerged - whether they chose to handle balls in order of arrival, prioritized certain colors, or considered how close each ball was to its destination.

The blindfolded round brings a whole new perspective to the discussion. Students can reflect on how they adapted their communication and relied on memory of their neighbors' positions. This demonstrates how networks must maintain reliable connections even when working with limited information.

Throughout these reflections, help students understand **how their individual simple decisions - called heuristics, simple rules that help make good choices quickly - contributed to solving a complex routing problem.**

By making straightforward choices at each step - like passing to the nearest person in the right direction - they collectively created effective pathways through the network. This mirrors how actual routing algorithms work, breaking down complex networking challenges into manageable decisions that lead to efficient solutions.



This practical demonstration helps pupils grasp how their simple decision-making process mirrors routing algorithms. By making choices about where to pass the ball, they have actually created and followed a simple algorithm, just like navigation apps do. Each student acted as a "node" in the network, making decisions based on simple rules:

- Choosing the best path option to pass the ball
- Managing multiple balls without dropping them (similar to how networks handle multiple data packets)
- Adapting their strategy when obstacles appeared

A routing algorithm can be explained to children as a set of instructions or rules that help us decide the best way to get from one place to another. Just like when they play a game and have to find the quickest way to reach the finish line, a routing algorithm does something similar. It looks at all the possible paths, considers different factors like distance, time, and obstacles, and then chooses the best path to reach the destination. In the world of computers, a routing algorithm is like a very smart map. It helps information travel through the Internet, which is a bit like a big city filled with roads. Sometimes, there's a lot of traffic or road work, so the routing algorithm helps the information find the best path. It thinks about things like how long it will take and how many other computers the information needs to visit along the way. And it updates itself if anything changes.

Explore additionally heuristic choices and routing with a board game (optional)

After exploring the routing algorithm concepts through the physical activities in the schoolyard, you can optionally use the popular board game "**Ticket to Ride**" to organise a last playful session and explore the concepts more quietly. The closest games library near you will likely have several versions of it to lend to you. You can try to contextualise it as the map collection is wide so children can relate to something they know - You can find maps of Europe, France, Germany, the UK, Italy, and The Netherlands.

You can find more information here: <https://www.daysofwonder.com/ticket-to-ride/maps-collection/>, https://boardgamegeek.com/wiki/page/Ticket_to_Ride_series.

Explain how players must plan routes for their trains, highlighting the link with the concept of routing in the itinerary. Play one or two rounds to familiarize students with the idea. It is not mandatory to play the game until the end, but it will give a nice touch of a gamified vision of a complex concept to your pupils. Bring back the classroom in a normal setting, ask students to think about how they planned their routes in the game if they have played it, and how this might relate to programming routing algorithms in real life. Encourage them to think about the factors they had to consider in the game, such as distance and obstacles, and how they might apply to designing a routing algorithm.

Preparation and teacher notes

Prepare documentation about transport modes based on your chosen research approach:

- Print pictures and descriptions of various transport types
- Provide transportation-related books
- Create a curated collection of printed resources for students to study
- Collect local transport maps and schedules as examples
- Prepare computers or tablets for online transportation research

Gather art supplies for creating the transportation panels (A3 paper or foam boards, markers, scissors, glue)

Prepare maps and cartography materials:

- Print maps of the local area
- Prepare overlays for different transport routes
- Create symbols for various obstacles and constraints

Build up - 1 hour

Orientation, at its core, is the ability to locate oneself in one's environment and navigate through it. Throughout human history, it has been a fundamental concept impacting our lives, routines, systems, and businesses. From ancient sailors using the stars to guide their way across the oceans, to modern-day drivers using GPS systems to navigate crowded city streets, orientation plays a crucial role in our everyday actions and decisions.

In terms of routines and systems, orientation helps us create efficient paths for our daily commutes. On a larger scale, it influences urban planning, transportation networks, and international trade routes. This concept, although often taken for granted, greatly shapes our physical world and how we interact with it.

The concept of routing algorithms can be seen as an exemplification of orientation. These algorithms, used in digital mapping tools and navigation systems, take various factors into account such as distance, time, and obstacles to determine the most efficient path from one location to another. In essence, they perform the same function that our brains do when we mentally map out our route to a destination.

Building upon the "Pass the Parcel" activity where students acted as network nodes making routing decisions, this activity expands that understanding into real-world navigation. Just as students learned to make simple routing choices when passing balls through their human network, they will now apply similar decision-making processes i.e. heuristics to transportation routes. Students will design and analyze routes based on different transportation systems and constraints, developing a deeper understanding of how routing algorithms work in practice. Through this connection between the abstract concept of network routing and real-world navigation, they will improve their geographical understanding, problem-solving skills, and grasp of algorithmic thinking.

Exploring and researching transportation systems

- During a first rapid roundtable, ask students to think about the different means of transport that exist and in which context they are used, by whom and how. You can do it with a simple roundtable or based on voluntary participation.
- Then, divide the students into small groups, to research the different modes of transport identified more deeply: their efficiency, their costs, who can use them and how, what are their level of accessibility, and their environmental impact. They should identify if they are used for daily activities or if they are more punctual, and how this impacts the environmental footprint of the transport choice. This phase aims to build knowledge on getting around for daily travels.
- Each group presents its findings to the class. Optionnaly, you can create an artistic commitment to the activity by asking each group to create a panel presenting the transport mode they have studied, which you can then hang in the classroom.

Crossing routing and orientation through map exploration

Just like we experienced in the schoolyard with our human network, a map also represents a network of possible paths. In our activity, each student was a "node" that had to make simple decisions to move the balls (our "data") toward their destination. Similarly, on a map, each intersection is a decision point where we must choose the best path. This approach helps us understand how navigation algorithms work in real life, whether we are finding our way on foot, by bicycle, by car, or by public transport.

Divide again the classroom into small groups (you can reuse the same groups as in the step before). Each group is attributed with a transport system basically: **pedestrian / bicycle / personal car / public bus**.

Distribute a set of maps to all groups: pupils will work with two different scenarios. First, "**the easy way**" uses a basic street map with minimal obstacles. Provide then "**the challenging way**" by adding complexity information. You can use pictograms under your map to provide additional information to consider such as: traffic congestion, construction work, slippery paths or transport-specific limitations.

Each group will create a routing algorithm for their assigned transport mode (walking, cycling, driving, or taking the bus). Your algorithm should find the best path from point A to point B while considering multiple factors.

- For **pedestrians**, think about safety with proper sidewalks and crossings, accessibility for people with reduced mobility, and overall distance and time constraints.
- For **bicycles**, consider available bike lanes, road safety by avoiding major roads and highways, and terrain difficulty like hills.
- For **personal cars**, account for traffic conditions, parking availability, and potential detours due to one-way streets.
- For **public buses**, examine bus stop locations, how often buses run, whether routes go directly to the destination, and how the bus schedule might affect travel time

Teacher notes: creating the maps

For creating the maps, several digital mapping tools are available:

Google Maps is an excellent starting point, offering a user-friendly interface and familiar features. You can capture screenshots of the basic map view and enhance them with important overlays such as bicycle paths, real-time traffic data, public transport routes, and bus stop locations.

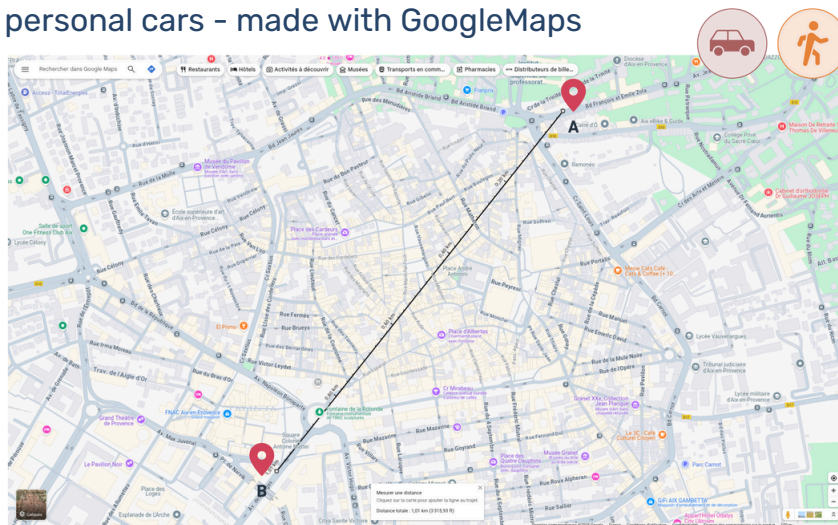
For more specialized and detailed mapping needs, consider these alternatives:

- **OpenStreetMap** (openstreetmap.org) provides open-source maps with detailed layers (including public transport)
- **CycloSM** (cyclosm.org) is designed for cycling-related navigation with detailed bicycle infrastructure
- **MilvusMap** (milvusmap.eu) offers detailed information including accessibility related layers and the possibility to print the maps in PDF.

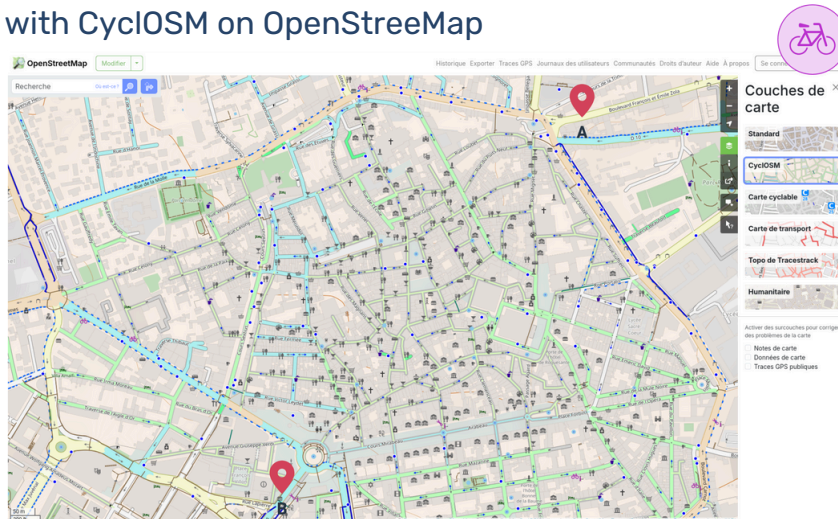
You can also explore the OpenStreetMap database of tools for maps on paper: https://wiki.openstreetmap.org/wiki/OSM_on_Paper

Example of maps generated with several tool for the “Easy Way” - Aix-en-Provence, France

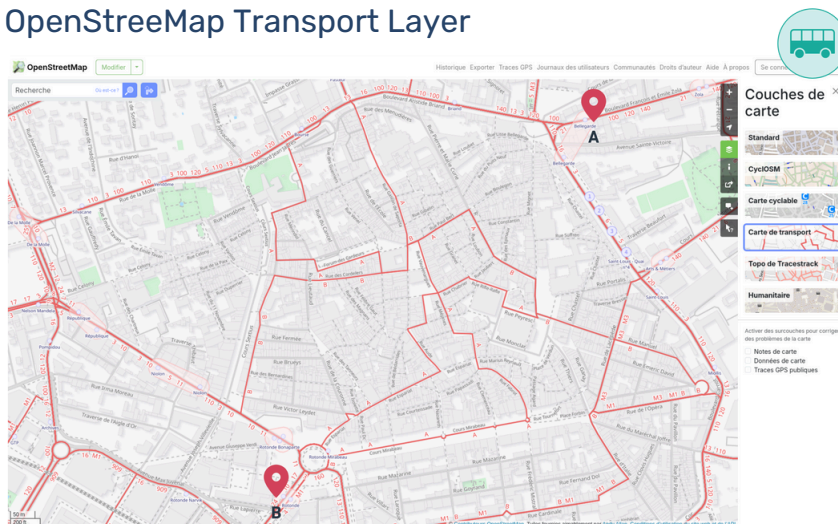
1. Map with no specific information - Pedestrian and personal cars - made with GoogleMaps



2. Map with soft mobility information - Bicycle - made with CycloSM on OpenStreetMap

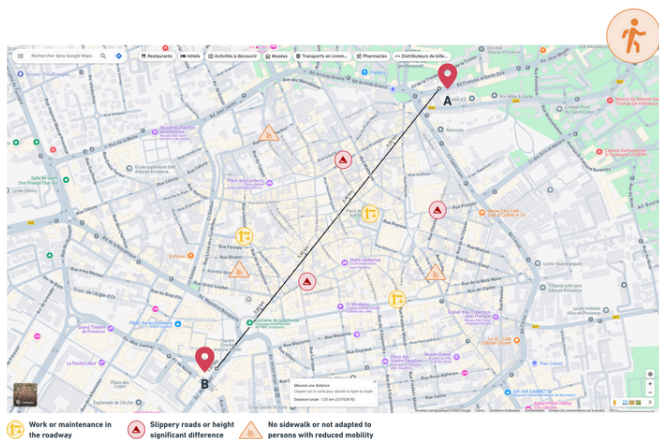


3. Map with bus stops - Public Transport - made with OpenStreetMap Transport Layer

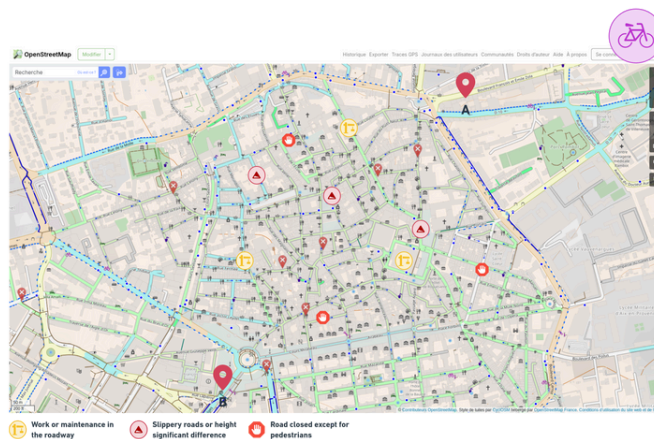


Example of maps generated with several tool for the “Challenging Way” - Aix-en-Provence, France

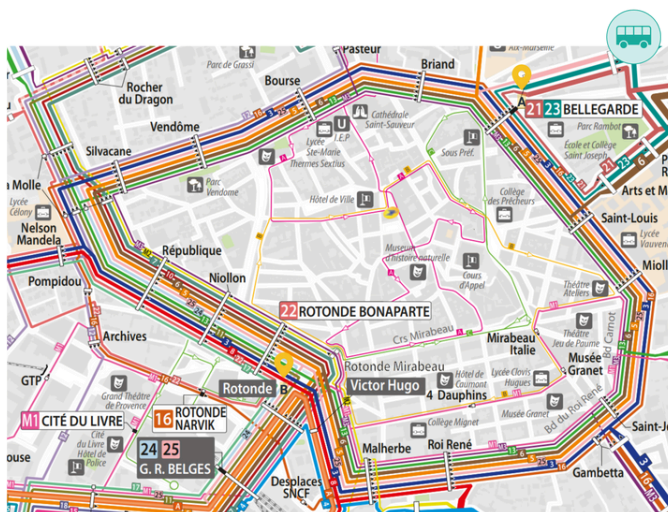
1. Map with information regarding sidewalks and accessibility - Pedestrian - made with GoogleMaps + icons



2. Map with road surface conditions - Bicycle - made with CycloSM + Map note layer on OpenStreetMap + icons

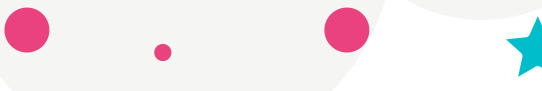


3. Map with bus detailed information - downloaded from the local public transport company maps



4. Map with traffic information in real time - Personal Cars - made with GoogleMaps





Clean-up - 15 minutes

At the end of the activity, gather the class for a thoughtful discussion about their experience with routing algorithms. Start by asking students to share how their routing decisions evolved when faced with different constraints and what main challenges they encountered for each transport mode.

Guide them to reflect on how they handled unexpected obstacles along the way. Encourage students to compare the different transport modes they explored, considering which routes proved fastest, safest, or most environmentally friendly.

This discussion should highlight how different constraints affected each transport mode in unique ways, leading to varied optimal solutions.

Ask pupils to analyze the heuristics they used when planning routes for instance:

- **"Always choose the straightest path toward the destination"**
- **"Avoid busy streets during peak hours"**
- **"Select routes with dedicated lanes for their transport mode"**
- **"Choose paths with the fewest turns or intersections"**

Help students understand that while these simple rules do not always lead to the absolute best solution, they often help us make good decisions quickly. This mirrors how actual routing algorithms work - they use similar heuristics to balance finding a good route with computational efficiency.

Ask students to compare the effectiveness of different heuristics. For instance, did "choosing the straightest path" work better for cars than for pedestrians? Did "avoiding busy streets" matter more during certain times of day? This discussion helps reinforce how different constraints and priorities affect route planning.



Preparation and teacher notes

Prepare safety and logistics requirements

- Obtain necessary permissions for the field trip
- Arrange for additional adult supervision if needed
- Prepare emergency contact information and first aid kit
- Check weather conditions before the expedition

Prepare navigation tools for each group:

- Compasses and instruction guides for compass use
- Mobile devices with GPS or autonomous GPS
- Printed maps of the area
- List of relevant street signs and landmarks

Plan timing and logistics:

- Map out estimated duration for each route
- Identify meeting points along the way
- Plan for rest stops and water breaks
- Arrange transportation back to school if needed

Optional Rehearsal phase - Plan an expedition to test the algorithms on a real itinerary - Half-day

As an additional stage, and if conditions allow it (logistically, in terms of time or supervision), an engaging conclusion to the lesson would be to plan an expedition that allows students to apply routing algorithms outside the classroom in real life. Choose a destination for the route. It should be a place that is relevant to the pupils and offers a variety of possible routes and intersections.

A relevant example for your pupils could be to choose the route **from their current primary school to the secondary school most of them will attend the following years**. This not only applies their learning in a real-world context but also helps them prepare for a significant upcoming change in their lives.

To test different decision-making choices, you can create groups and provide each of them with a navigation constraints: one can have a compass, another one a GPS and the last one can have the constraint to only follow street signs.

- **Compass constraints and interest:** Using a compass in this activity presents an interesting challenge. Unlike modern digital mapping tools, a compass does not provide a pre-determined route or indicate the direction of the destination. Instead, the compass points to the North, requiring students to orient themselves based on their understanding of geographical directions and their location relative to the destination. The compass will guide them in choosing between two routes options. However, this choice may not always be straightforward. They might find that the direction chosen based on the compass leads to a closed road or a one-way street for instance that they will enlarge the travel. These real-world constraints will require them to constantly reassess and refine their choices, providing a deeper understanding of routing and the complexities of real-world navigation.

- **GPS constraints and interest:** Using GPS navigation provides students with precise location data and turn-by-turn directions. This tool offers real-time positioning and multiple route options. However, the challenge lies in understanding how the GPS calculates these routes and what factors it considers. Students will need to analyze why the GPS suggests certain paths over others, considering factors like traffic, distance, and road types. This helps them understand how digital routing algorithms work in practice and how they balance different variables to determine optimal routes.
- **Street signs constraints and interest:** Following only street signs presents a unique navigation challenge that mirrors historical wayfinding methods. Students must rely on physical infrastructure and urban planning to reach their destination. This approach requires careful attention to street names, direction signs, and local landmarks. It demonstrates how cities are designed to be navigable without digital assistance and helps students develop spatial awareness. The limitation of using only street signs may lead to longer routes or temporary confusion, highlighting the evolution of navigation methods and the importance of clear urban signage.

Once the activity is completed, engage in a discussion with the students. Talk about the route, the obstacles they encountered, and how they overcame them. Reflect on how the experience compared to their planned routes, and how they could improve their routing algorithm based on this real-world experience.

Optionnal additional clean-up - 15 minutes

After completing the expedition, gather the class for a final discussion to analyze the experience and draw parallels between all three phases of the activity:

- **In the unplugged ball game, students learned basic routing concepts by making simple decisions to pass "data" through a human network**
- **During map work, they applied these concepts to create theoretical algorithms considering various transport constraints**
- **Finally, through the field trip, they tested these algorithms in real-world conditions**

Compare how their theoretical algorithms performed in practice and discuss unexpected challenges they encountered. This three-step progression from abstract concept to real-world application helps reinforce the connection between algorithmic thinking and orientation.

To extend the learning experience, consider creating a collaborative digital map marking the discovered routes, obstacles, and points of interest. The pupils could also analyze the environmental impact of different transport choices by calculating carbon footprints for each route, or explore how their routes might change during different seasons or weather conditions.